CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 5: Arithmetic Progressions

Every formula and result of the chapter on one sheet

1	General form of an AP	$a,\ a+d,\ a+2d,\ a+3d,\ \ldots$
2	Common difference	$d = a_n - a_{n-1}$
3	nth term	$a_n = a + (n-1)d$
4	nth term from the end	$l - (n-1)d$
5	Number of terms	$n = \frac{l-a}{d} + 1$
6	Sum of the first n terms	$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$
7	Sum when the last term is known	$S_n = \frac{n}{2}(a+l)$
8	nth term from the sums	$a_n = S_n - S_{n-1}$
9	Sum of the first n positive integers	$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$
10	Sum of the first n odd numbers	$1 + 3 + 5 + \cdots + (2n-1) = n^2$
11	Sum of the first n even numbers	$2 + 4 + \cdots + 2n = n(n+1)$
12	Three terms in AP	$a-d,\ a,\ a+d$
13	Condition for an AP	a, b, c are in AP if and only if $2b = a + c$
14	Arithmetic mean of a and c	$b = \frac{a+c}{2}$